

Spatial Autocorrelation: A Review

John Smith, Jane Doe

1. Introduction

Spatial autocorrelation measures the degree to which a set of spatial features and their associated data values are related to each other.

2. Methods

Morans I is the most widely used measure of global spatial autocorrelation. It compares the value of a variable at one location to the average value of the same variable at all other locations.

Gearys C is another measure that is more sensitive to local spatial autocorrelation patterns. Unlike Morans I, Gearys C is a ratio of the variance of the variable to the variance of the residuals.

3. Applications in Smart Cities

In smart city research, spatial autocorrelation analysis has been applied to POI distribution patterns, traffic flow patterns, and land use patterns.